

ME 2261 - Embedded Systems Log Book**Lab Practical 3****Task 1**

$$5 + 7 + 8 = R1$$

```
1 LDI R16, 5
2 LDI R17, 7
3 ADD R16, R17
4 LDI R17, 8
5 ADD R16, R17
6 MOV R1, R16
7 NOP
8
```

Processor		
Name	Value	
Program Counter	0x00000000	
Stack Pointer	0x08FF	
X Register	0x0000	
Y Register	0x0000	
Z Register	0x0000	
Status Register	I T H S V N Z C	
Cycle Counter	0	
Frequency	1.000 MHz	
Stop Watch	0.00 µs	
Registers		
R00	0x00	
R01	0x00	
R02	0x00	
R03	0x00	
R04	0x00	
R05	0x00	
R06	0x00	
R07	0x00	
R08	0x00	
R09	0x00	
R10	0x00	
R11	0x00	
R12	0x00	
R13	0x00	
R14	0x00	
R15	0x00	
R16	0x00	
R17	0x00	
R18	0x00	
R19	0x00	
R20	0x00	

```

1 LDI R16, 5
2 LDI R17, 7
3 ADD R16, R17
4 LDI R17, 8
5 ADD R16, R17
6 MOV R1, R16
7 NOP
8

```

Processor	
Name	Value
Program Counter	0x00000001
Stack Pointer	0x08FF
X Register	0x0000
Y Register	0x0000
Z Register	0x0000
Status Register	☐ T ☐ H ☐ S ☐ V ☐ N ☐ Z ☐ C
Cycle Counter	0
Frequency	1.000 MHz
Stop Watch	0.00 µs
Registers	
R00	0x00
R01	0x00
R02	0x00
R03	0x00
R04	0x00
R05	0x00
R06	0x00
R07	0x00
R08	0x00
R09	0x00
R10	0x00
R11	0x00
R12	0x00
R13	0x00
R14	0x00
R15	0x00
R16	0x05
R17	0x00
R18	0x00
R19	0x00
R20	0x00

```

1 LDI R16, 5
2 LDI R17, 7
3 ADD R16, R17
4 LDI R17, 8
5 ADD R16, R17
6 MOV R1, R16
7 NOP
8

```

Processor	
Name	Value
Program Counter	0x00000002
Stack Pointer	0x08FF
X Register	0x0000
Y Register	0x0000
Z Register	0x0000
Status Register	ITHSVNZC
Cycle Counter	1
Frequency	1.000 MHz
Stop Watch	1.00 µs
Registers	
R00	0x00
R01	0x00
R02	0x00
R03	0x00
R04	0x00
R05	0x00
R06	0x00
R07	0x00
R08	0x00
R09	0x00
R10	0x00
R11	0x00
R12	0x00
R13	0x00
R14	0x00
R15	0x00
R16	0x05
R17	0x07
R18	0x00
R19	0x00
R20	0x00

```

1  LDI R16, 5
2  LDI R17, 7
3  ADD R16, R17
4  LDI R17, 8
5  ADD R16, R17
6  MOV R1, R16
7  NOP
8

```

Processor	
Name	Value
Program Counter	0x00000003
Stack Pointer	0x08FF
X Register	0x0000
Y Register	0x0000
Z Register	0x0000
Status Register	I T H S V N Z C
Cycle Counter	2
Frequency	1.000 MHz
Stop Watch	2.00 µs
Registers	
R00	0x00
R01	0x00
R02	0x00
R03	0x00
R04	0x00
R05	0x00
R06	0x00
R07	0x00
R08	0x00
R09	0x00
R10	0x00
R11	0x00
R12	0x00
R13	0x00
R14	0x00
R15	0x00
R16	0x0C
R17	0x07
R18	0x00
R19	0x00
R20	0x00

```

1 LDI R16, 5
2 LDI R17, 7
3 ADD R16, R17
4 LDI R17, 8
5 ADD R16, R17
6 MOV R1, R16
7 NOP
8

```

Processor	
Name	Value
Program Counter	0x00000004
Stack Pointer	0x08FF
X Register	0x0000
Y Register	0x0000
Z Register	0x0000
Status Register	I T H S V N Z C
Cycle Counter	3
Frequency	1.000 MHz
Stop Watch	3.00 µs
Registers	
R00	0x00
R01	0x00
R02	0x00
R03	0x00
R04	0x00
R05	0x00
R06	0x00
R07	0x00
R08	0x00
R09	0x00
R10	0x00
R11	0x00
R12	0x00
R13	0x00
R14	0x00
R15	0x00
R16	0x0C
R17	0x08
R18	0x00
R19	0x00
R20	0x00

```

1  LDI R16, 5
2  LDI R17, 7
3  ADD R16, R17
4  LDI R17, 8
5  ADD R16, R17
6  MOV R1, R16
7  NOP
8

```

Processor	
Name	Value
Program Counter	0x00000005
Stack Pointer	0x08FF
X Register	0x0000
Y Register	0x0000
Z Register	0x0000
Status Register	☐ T ☒ H ☐ S ☐ V ☐ N ☐ Z ☐ C
Cycle Counter	4
Frequency	1.000 MHz
Stop Watch	4.00 µs
Registers	
R00	0x00
R01	0x00
R02	0x00
R03	0x00
R04	0x00
R05	0x00
R06	0x00
R07	0x00
R08	0x00
R09	0x00
R10	0x00
R11	0x00
R12	0x00
R13	0x00
R14	0x00
R15	0x00
R16	0x14
R17	0x08
R18	0x00
R19	0x00
R20	0x00

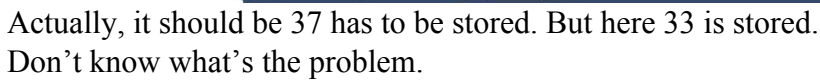
```

1  LDI R16, 5
2  LDI R17, 7
3  ADD R16, R17
4  LDI R17, 8
5  ADD R16, R17
6  MOV R1, R16
7  NOP
8

```

Processor	
Name	Value
Program Counter	0x00000006
Stack Pointer	0x08FF
X Register	0x0000
Y Register	0x0000
Z Register	0x0000
Status Register	I T H S V N Z C
Cycle Counter	5
Frequency	1.000 MHz
Stop Watch	5.00 µs
Registers	
R00	0x00
R01	0x14
R02	0x00
R03	0x00
R04	0x00
R05	0x00
R06	0x00
R07	0x00
R08	0x00
R09	0x00
R10	0x00
R11	0x00
R12	0x00
R13	0x00
R14	0x00
R15	0x00
R16	0x14
R17	0x08
R18	0x00
R19	0x00
R20	0x00

Task 3



```

1 LDI R20,0x75
2 STS 0x80,R20
3 LDS R20,0x80
4 STS 0x81,R20
5 NOP
6

```

Memory 1		Memory: data REGISTERS		Address: 0x0081,data	
data 0x0081	51	00	00	00	00
data 0x0096	00	00	00	00	00
data 0x00AB	00	00	00	00	00
data 0x00C0	20	00	06	00	00
data 0x00D5	00	00	00	00	00
data 0x00EA	00	00	00	00	00
data 0x00FF	00	00	00	00	00
data 0x0114	00	00	00	00	00
data 0x0129	00	00	00	00	00
data 0x013E	00	00	00	00	00
data 0x0153	00	00	00	00	00
data 0x0168	00	00	00	00	00
data 0x017D	00	00	00	00	00
data 0x0192	00	00	00	00	00
data 0x01A7	00	00	00	00	00
data 0x01BC	00	00	00	00	00
data 0x01D1	00	00	00	00	00
data 0x01E6	00	00	00	00	00
data 0x01FB	00	00	00	00	00
data 0x0210	00	00	00	00	00
data 0x0225	00	00	00	00	00
data 0x023A	00	00	00	00	00
data 0x024F	00	00	00	00	00
data 0x0264	00	00	00	00	00
data 0x0279	00	00	00	00	00
data 0x028E	00	00	00	00	00
data 0x02A3	00	00	00	00	00
data 0x02B8	00	00	00	00	00
data 0x02CD	00	00	00	00	00
data 0x02E2	00	00	00	00	00
data 0x02F7	00	00	00	00	00
data 0x030C	00	00	00	00	00
data 0x0321	00	00	00	00	00
data 0x0336	00	00	00	00	00
data 0x034B	00	00	00	00	00
data 0x0360	00	00	00	00	00
data 0x0375	00	00	00	00	00
data 0x038A	00	00	00	00	00
data 0x039F	00	00	00	00	00

Same problem

The value of Hexa Decimal 51 is stored.

Sir suggested, there could be a software problem.